Thermodynamics

MAIS Physics & Astronomy Club (PAC) · Olympiad lecture + problem handout

8 lecture sections · 15 problems · 1 interactive experiment · 2 vector figures

Start with a physical picture, then introduce only the mathematics needed to make it predictive. The lecture connects everyday reasoning to astronomical measurements. Two vector diagrams and an interactive experiment support the argument; 15 graded problems and independent worked solutions turn the ideas into practice.

▼ Contents & learning path

1. Temperature is a state; heat is a transfer
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1. Temperature is a state; heat is a transfer

A gas has a temperature. It does not "contain heat" in the same sense. Heat is energy crossing a boundary because of a temperature difference; work is another way to transfer energy. Thermal equilibrium means no net heat flow between systems in contact.

For a dilute ideal gas,

$$PV = Nk_BT = nRT. \tag{1}$$

Use kelvins. Here N counts particles, n counts moles, $k_B = 1.380649 \times 10^{-23}\ \mathrm{J\,K^{-1}}$ and $R = 8.314462618\ \mathrm{J\,mol^{-1}\,K^{-1}}$. The model neglects molecular volume and long-range interactions; it is a useful approximation, not a universal law.

2. The first law is an energy account

We take $Q > 0$ when heat enters the gas and $W > 0$ when the gas does work on its surroundings:

$$\Delta U = Q - W, \qquad W = \int P\, dV. \tag{2}$$

For a monatomic ideal gas, $U = \frac{3}{2} nRT$. Internal energy depends only on temperature; heat and work depend on the path. A P–V diagram makes this visible: work is the signed area under the curve.

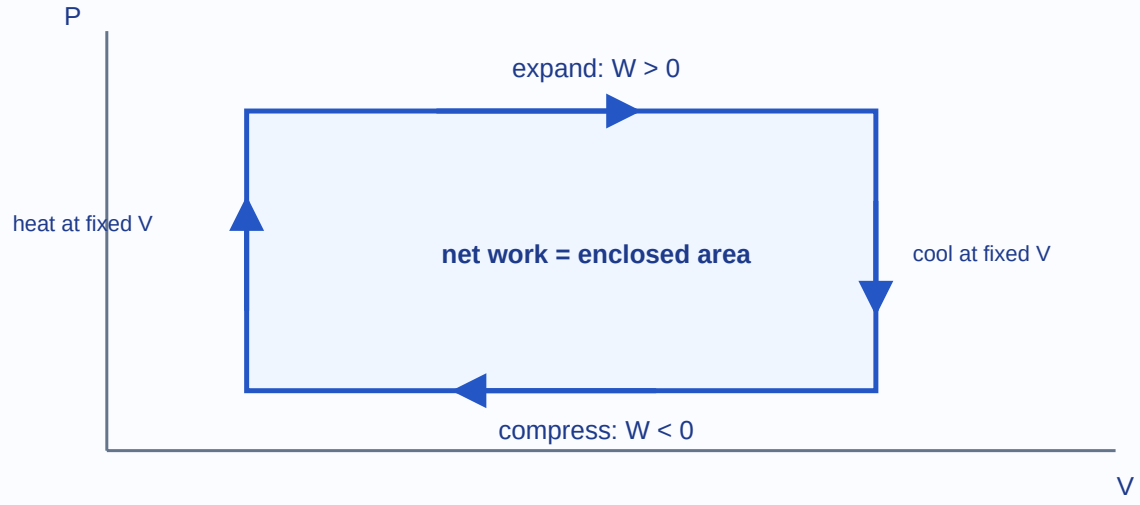


Figure 5. Pressure and volume describe the state; heat and work describe energy transfer along a process. The sign convention is ΔU = Q − W, with W done by the gas.

3. Four processes, four energy bills

Process	Constraint	Ideal-gas consequence
Isochoric	V fixed	$W = 0$, $Q = \Delta U$
Isobaric	P fixed	$W = P\Delta V = nR\Delta T$
Isothermal	T fixed	$\Delta U = 0$, $Q = W = nRT \ln(V_2/V_1)$
Reversible adiabatic	$Q = 0$	PV^γ fixed, $TV^{\gamma-1}$ fixed

For a monatomic gas, $C_V = 3R/2$, $C_P = 5R/2$ and $\gamma = 5/3$ per mole. At constant pressure, added heat must both raise internal energy and pay for expansion work.

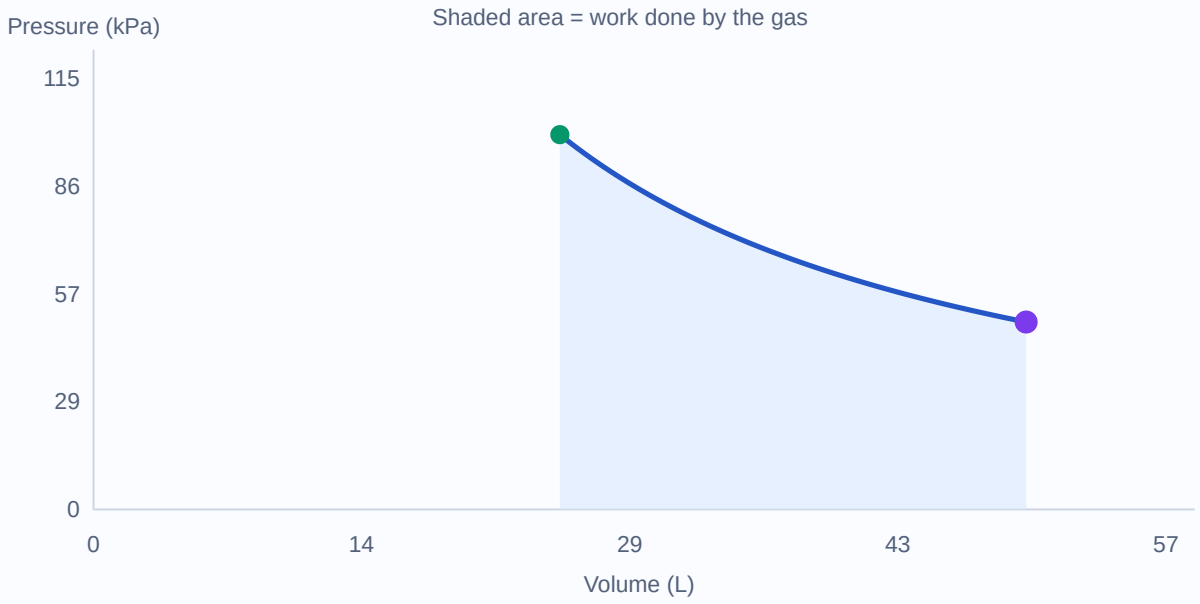
INTERACTIVE EXPERIMENT

Same starting state. Different energy bills.

One mole of monatomic ideal gas starts at 300 K and 100 kPa. The slider sets V_2/V_1, except for the isochoric process where it sets P_2/P_1. W is work done by the gas; Q = ΔU + W.

Final T	Final P	Work W
300.0 K	**50.00 kPa**	**1728.9 J**

Heat Q	ΔU	Final V
1728.9 J	**0.0 J**	**49.89 L**



First-law check: Q − W = 0.0 J, equal to ΔU = 0.0 J. Monatomic gas: $C_v = 3R/2$, $\gamma = 5/3$.

Try this. Compare isothermal and adiabatic expansion to twice the volume. Which needs heat input? Which cools? Check every result against the first law.

Snapshot of the experiment. Open the HTML edition to change the controls.

4. Cycles and engines

Over a complete cycle, the gas returns to its initial state, so $\Delta U = 0$ and net heat equals net work. A clockwise loop on a P–V diagram does positive work. Efficiency is useful work divided by **heat input**, not by the net heat:

$$\eta = \frac{W_{\text{net}}}{Q_{\text{in}}}. \tag{3}$$

Identify which legs absorb heat before computing efficiency. The Carnot limit $1 - T_c/T_h$ applies to a reversible engine between two reservoirs, not to every arbitrary cycle.

5. A planet's energy balance

A planet intercepts starlight over a disk πR^2 and, in a uniform-temperature model, emits over $4\pi R^2$. With incident flux F and Bond albedo A,

$$(1 - A) F \pi R^2 = 4\pi R^2 \sigma T_{\text{eff}}^4. \tag{4}$$

The radius cancels. The factor of four is geometry. The effective radiating temperature need not equal the surface temperature: an atmosphere can change where radiation escapes.

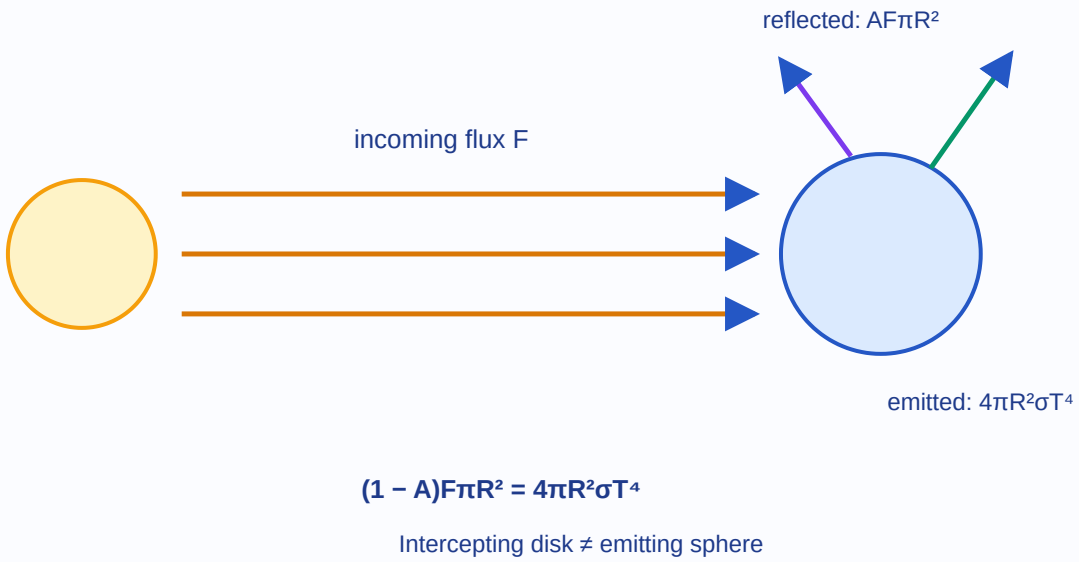


Figure 6. The factor of four is geometric. Effective radiating temperature need not equal surface temperature when an atmosphere is present.

6. Three ways energy moves

Conduction transfers energy through microscopic interactions, with Fourier's law $\dot{Q} = kA\Delta T/\ell$ for a uniform slab. Convection carries energy with moving fluid; a rising parcel cools as pressure falls. Radiation transfers energy by photons and works through vacuum. In stars, opacity and temperature gradients help determine whether radiation or convection dominates.

7. Atmospheric and stellar scales

For an isothermal ideal atmosphere in constant gravity, hydrostatic balance gives

$$P(z) = P_0 e^{-z/H}, \qquad H = \frac{k_B T}{\mu m_u g}. \tag{5}$$

A scale height is the distance for a factor-e pressure drop. For a dry ideal atmosphere, the adiabatic temperature gradient is approximately g/c_p. Comparing a lifted parcel with its surroundings diagnoses buoyant stability.

A thermal timescale compares an energy reservoir with a loss rate. The Kelvin–Helmholtz scale $GM^2/(RL)$ estimates how long gravitational contraction could power a luminosity. It is not the nuclear main-sequence lifetime.

8. Check the model, not only the arithmetic

Ask whether the gas is ideal, whether a process is reversible, whether temperature is uniform, and which surface emits. Keep the first-law sign convention fixed. A numerical answer should pass an energy balance and a limiting-case check.

A useful habit. Before calculating, write the fixed quantity and the energy pathway. “No heat” does not mean “no temperature change”; adiabatic expansion cools a gas because it does work.

9. Lecture + Problems: predict, calculate, check

Fifteen original problems: **[1]** foundation, **[2]** multi-step Olympiad reasoning, **[3]** challenge. Use the constants supplied in the lecture unless a question overrides them. Explain assumptions and keep units. Every problem has a worked solution.

[1] Problem 1 — Read the state. One mole of ideal gas occupies 25 L at 100 kPa. Find its temperature and number of molecules.

Worked solution →

[1] Problem 2 — Isothermal expansion. One mole at 300 K expands reversibly to twice its volume. Find W, Q and ΔU.

Worked solution →

[2] Problem 3 — Constant-pressure heating. Heat one mole of monatomic gas from 300 to 400 K at constant pressure. Find W, ΔU and Q.

Worked solution →

[2] Problem 4 — Expansion without heat. One mole of monatomic gas at 300 K expands reversibly and adiabatically to twice its volume. Find final temperature and work.

Worked solution →

[1] Problem 5 — A rigid vessel. One mole of monatomic gas at 300 K is heated at fixed volume until its pressure doubles. Find final temperature and heat input.

Worked solution →

[3] Problem 6 — A rectangular engine. One mole of monatomic gas follows a clockwise rectangle: expand at $P = 300$ kPa from 10 to 30 L; cool at 30 L to 100 kPa; compress at 100 kPa to 10 L; heat at 10 L back to 300 kPa. Find net work, total heat input and efficiency.

Worked solution →

[2] Problem 7 — A bare planet. Incident flux is 1361 W/m². Find equilibrium effective temperature for albedo 0.3 and for 0.5, assuming uniform emission. If only the day side emitted with the same uniform dayside temperature, what would the first temperature become?

Worked solution →

[3] Problem 8 — One ideal greenhouse layer. An atmosphere is transparent to incoming starlight and absorbs all surface infrared radiation. It emits equally upward and downward as a blackbody. Show that the surface temperature is $2^{1/4}$ times the effective temperature. Evaluate for $T_{\text{eff}} = 254.6$ K and state the model's limits.

Worked solution →

[1] Problem 9 — Mixing water. Mix equal masses of water at 20°C and 80°C in an insulated container with negligible container heat capacity and the same specific heat throughout. Find the final temperature.

Worked solution →

[2] Problem 10 — Heat through a wall. A slab has $k = 0.20\ \text{W m}^{-1}\ \text{K}^{-1}$, area 0.010 m², thickness 5 cm and a 100 K temperature difference. Find the steady conductive power.

Worked solution →

[2] Problem 11 — An atmospheric ruler. For $T = 250$ K, mean molecular weight 28.97 and $g = 9.81\ \mathrm{m\,s^{-2}}$, find the isothermal scale height. What is the pressure one scale height above the surface relative to the surface?

Worked solution →

[2] Problem 12 — Could contraction power the Sun? Estimate $GM_\odot^2/(R_\odot L_\odot)$ in years. Why does this not give the Sun's main-sequence age?

Worked solution →

[3] Problem 13 — A photon random walk. In a crude stellar-interior model, photons travel mean free path $\ell = 1$ cm across a distance $L = 7 \times 10^8$ m. Estimate the escape time using $N \sim (L/\ell)^2$ steps. Why is this only an order-of-magnitude picture?

Worked solution →

[3] Problem 14 — Is the atmosphere stable? A dry parcel starts at 300 K and rises 1 km. Take $g = 9.81\ \mathrm{m\,s^{-2}}$, $c_p = 1004\ \mathrm{J\,kg^{-1}\,K^{-1}}$ and environmental lapse rate 6.5 K/km. Find the parcel and environment temperatures and decide whether the parcel is buoyant. Assume ideal gas and equal pressure at the new height.

Worked solution →

[3] Problem 15 — A blackbody dust grain. A spherical blackbody grain at distance r from a star absorbs over its cross-section and emits over its surface. Show $T_d = T_* \sqrt{R_*/(2r)}$. For $T_* = 5772$ K and $R_* = R_\odot$, find the temperature at 1 AU and the distance for 150 K.

Worked solution →

Further practice and provenance

The 15 exercises above and their solutions are **original MAIS PAC problems**, inspired by the skills in this syllabus topic. They are not copied from the papers below. The earlier club lecture compiled these official past-paper identifiers for further practice; only identifiers are reproduced, not copyrighted question statements or official solutions. Round and question labels retain the earlier compilation's notation and should be checked against the official paper.

- IOAA 2007 · TH · 1.9
- IOAA 2007 · TH · Q2
- IOAA 2008 · TH · S4

- IOAA 2008 · TH · S9
- IOAA 2010 · TH · S5
- IOAA 2010 · TH · S12
- IOAA 2012 · TH · S7
- IOAA 2013 · TH · S1
- IOAA 2013 · TH · S3
- IOAA 2014 · TH · P7
- IOAA 2014 · TH · P8
- IOAA 2014 · TH · P10
- IOAA 2014 · DA · D2
- IOAA 2015 · TH · L3
- IOAA 2015 · TH · S9
- IOAA 2016 · TH · T2
- IOAA 2019 · TH · 5
- IOAA 2020 · TH · 2
- IOAA 2021 · TH · Q2
- IOAA 2022 · TH · 5
- IOAA 2022 · TH · 11
- IOAA 2023 · TH · Q4

Sources and scope. Topic coverage follows the IOAA syllabus. Official problems and constants are available through the IOAA past-paper archive. The derivations, diagrams, experiments and numerical teaching examples in this handout were prepared independently for MAIS PAC. Calculator-level methods are emphasized; optional calculus connections are explicitly identified.